

Первообразная

$$F' = f$$

$$(F + C)' = f$$

$$\begin{aligned} F' &= f \\ G' &= f \end{aligned} \Rightarrow F = G + C$$

$$\int f dx = F + C. \quad (\text{совокупность функций - первообразных})$$

$$x^a' = a x^{a-1}$$

$$\sin x' = \cos x$$

$$\sinh x' = \cosh x$$

$$a^x' = \ln a \cdot a^x$$

$$\log_a x' = \frac{1}{x \cdot \ln a}$$

$$\operatorname{tg} x' = \frac{1}{\cos^2 x}$$

$$\operatorname{ctgh} x' = -\frac{1}{\sinh^2 x}$$

$$\arcsin' = \frac{1}{\sqrt{1-x^2}}$$

$$\operatorname{arctg} x' = \frac{1}{1+x^2}$$

Таблица

Дукама + m



$$1) \int x^{\alpha} dx = \frac{1}{\alpha+1} x^{\alpha+1} + C, \alpha \neq -1$$

$$2) \int \frac{1}{x} dx = \ln|x| + C$$

> 0

$\neq 0$

$$(\ln(-x))' = \frac{1}{-x} \cdot (-1) = \frac{1}{x}$$

$$3) \int \cos x dx = \sin x + C$$

$$4) \int \sin x dx = -\cos x + C$$

$$5) \int \sinh x dx = \cosh x + C$$

$$6) \int \cosh x dx = \sinh x + C$$

$$7) \int \frac{dx}{\cos^2 x} = \tan x + C$$

$$8) \int \frac{dx}{\sin^2 x} = -\cot x + C$$

$$9) \int \frac{dx}{\cosh^2 x} = \tanh x + C$$

$$10) \int \frac{dx}{\sinh^2 x} = -\coth x + C$$

$$11) \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C = -\arccos x + C.$$

$$12) \int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arcsinh} x + C$$

$$13) \int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C$$

$$14) \int \frac{dx}{1+x^2} = \arctan x + C = -\operatorname{arccot} x + C.$$

$$15) \int a^x dx = \frac{a^x}{\ln a} + C.$$

Методы интегрирования

Замена переменных

$$\int f(u(t)) \overbrace{u'(t)}^{du} dt =$$

$$= \int f(u) du.$$

пример:

$$\int \sin x^2 \underbrace{2x dx}_{x^2 dx} = \int \sin u du = -\frac{\cos x^2}{2} + C.$$

$$\text{Cau} \quad u(t) = at + b \quad \text{Cau}$$

$$u' = a$$

$$\Rightarrow \int f(at+b) dt = \int f(u) \cdot \frac{u'}{a} dt =$$

$$= \frac{1}{a} \int f(u) du$$

$$\int \sin 2x dx = -\frac{1}{2} \cos 2x + C. \quad \begin{array}{l} t=2x; t'=2 \\ dt=2dx, \\ dx=\frac{dt}{2} \end{array}$$

πp. uuey:

$$\int \frac{dx}{1+2x^2} = \int \frac{dx}{1+(\sqrt{2}x)^2} = \frac{1}{\sqrt{2}} \arctg \sqrt{2}x + C.$$

$$- \int \operatorname{tg} x dx = \int \frac{\sin x}{\cos x} dx, \quad \begin{array}{l} \cos x = t \\ -\sin x dx = dt \end{array}$$

$$= \int -\frac{dt}{t} = -\ln|t| + C = \boxed{-\ln|\cos x| + C.}$$

$$- \int \frac{x}{\sqrt{1-x^2}} dx = \left| \begin{array}{l} u = 1-x^2, \\ du = -2x dx \end{array} \right| = -\frac{1}{2} \int \frac{du}{\sqrt{u}} =$$

$$= -\frac{1}{2} \cdot (-\frac{1}{\frac{1}{2}}) = \frac{1}{2} \cdot \frac{1}{\sqrt{u}} = \frac{1}{2\sqrt{1-x^2}} + C.$$

$$= -\frac{1}{2} \int u^{-\frac{1}{2}} du = -\frac{1}{2} \cdot \frac{u^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C = -\frac{1}{2} \cdot \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C = -\sqrt{1-x^2} + C.$$

$$\int \frac{x}{\sqrt{1-x^2}} dx = \left| \begin{array}{l} x = \sin t \\ dx = \cos t dt \end{array} \right| = \int \frac{\sin t \cdot \cos t dt}{\sqrt{1-\sin^2 t}} = \int \sin t dt = -\cos t + C = -\sqrt{1-x^2} + C.$$

$$\int \frac{dx}{(1+x)\sqrt{x}} = \left| \begin{array}{l} \sqrt{x} = t, \\ \frac{1}{2\sqrt{x}} dx = dt \end{array} \right| = \int \frac{2 dt}{1+t^2} = 2 \arctan t + C.$$

$$\int \frac{dx}{\sqrt{1+x^2}} = \left| \begin{array}{l} y = x + \sqrt{1+x^2} \\ dy = \left(1 + \frac{x}{\sqrt{1+x^2}}\right) dx = \frac{y}{\sqrt{1+x^2}} dx \\ \frac{dy}{y} = \frac{dx}{\sqrt{1+x^2}} \end{array} \right| =$$

$$= \int \frac{dy}{y} = \ln|y| + C = \ln(x + \sqrt{1+x^2}) + C.$$

гиперболический
логарифм
↓

$$\int \frac{dx}{\sqrt{x^2-1}} = \ln |x + \sqrt{x^2-1}| + C.$$

$$\int \frac{dx}{1-x^2} = \frac{1}{2} \int \left(\frac{1}{1-x} + \frac{1}{1+x} \right) dx =$$

$$= \frac{1}{2} (\ln |1+x| - \ln |1-x|) + C =$$

$$= \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C \quad (\text{высокий логарифм})$$

$$\int \frac{dx}{1-2x^2} = \frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}x}{1-\sqrt{2}x} \right| + C.$$

$$\int \frac{dx}{\cosh x} = \left| \begin{array}{l} t = \cosh x \\ dt = \sinh x dx \end{array} \right| = \int \frac{dt}{t \sinh x} = \int \frac{dt}{t \sqrt{t^2-1}} =$$

$$= \int \frac{2dx}{e^x + e^{-x}} = \left| \begin{array}{l} t = e^x \\ dt = e^x dx \end{array} \right| = \int \frac{2dt}{t(t + \frac{1}{t})} = \int \frac{2dt}{t^2+1} =$$

$$= 2 \operatorname{arctg} e^x + C.$$